

National Institute of Technology, Tiruchirappalli

Department of Physics

Date: 15/10/2025 & Time: 9.00 am-10.00 am CT-I

Subject: Physics-I (PHIR11)

Branch and Section: ICE-B

Answer all Questions

20 Marks

- ✓ 1. State the necessary condition for stimulated emission and derive the relation between Einstein's "A" and "B" coefficients. (4)
- ✓ 2. A 10 mW He-Ne laser has efficiency of 1%. Assume that all input energy is utilized in pumping the atoms from the ground state to the excited state, which is 20 eV above the ground state. Find how many atoms are promoted to the excited state in one second. (2)
- ✓ 3. Explain in brief three and four level pumping scheme. Why four level scheme is preferred over three level schemes? (3)
- ✓ 4. (a) Explain the phenomenon of total internal reflection of light in optical fibers.
(b) Describe the propagation of light in optical fiber and obtain expression for numerical aperture. (4)
- ✓ 5. In an optical fiber, the core material has refractive index 1.6 and refractive index of clad material is 1.3. What is the value of critical angle? Also calculate the value of angle of acceptance cone. (2)

Choose the correct answer (Each carries 1 mark)

(5)

6. If the frequency of emitted photon is 10 Hz, the ratio of Einstein's coefficient is
a) 2.177×10^{-41} b) 3.177×10^{-41} c) 5.177×10^{-51} d) 6.177×10^{-51}
7. Which method is used by the curiosity rover's ChemCam lasers to find the chemical composition?
a) Atomic absorption laser spectroscopy
b) Laser-induced breakdown spectroscopy
c) UV-visible spectroscopy
d) Energy dispersive spectroscopy
8. Which of the following substances would have the highest optical density?
a) Air b) Ruby c) Crown glass d) Sapphire
9. A diode laser having a bandgap of 2.32×10^{-19} J shall have emission in _____ of EM spectrum.
a) Far-IR region b) UV region c) Visible region d) Near-IR region
10. Which of the following loss occurs inside the fiber optics?
a) Radiative loss b) Scattering c) Absorption d) Attenuation

[Constants: $k_B = 1.38 \times 10^{-23}$ J/K; $e = 1.602 \times 10^{-19}$ C; $h = 6.626 \times 10^{-34}$ Js; $c = 3 \times 10^8$ m/s; $1\text{eV} = 1.60218 \times 10^{-19}$ J]
